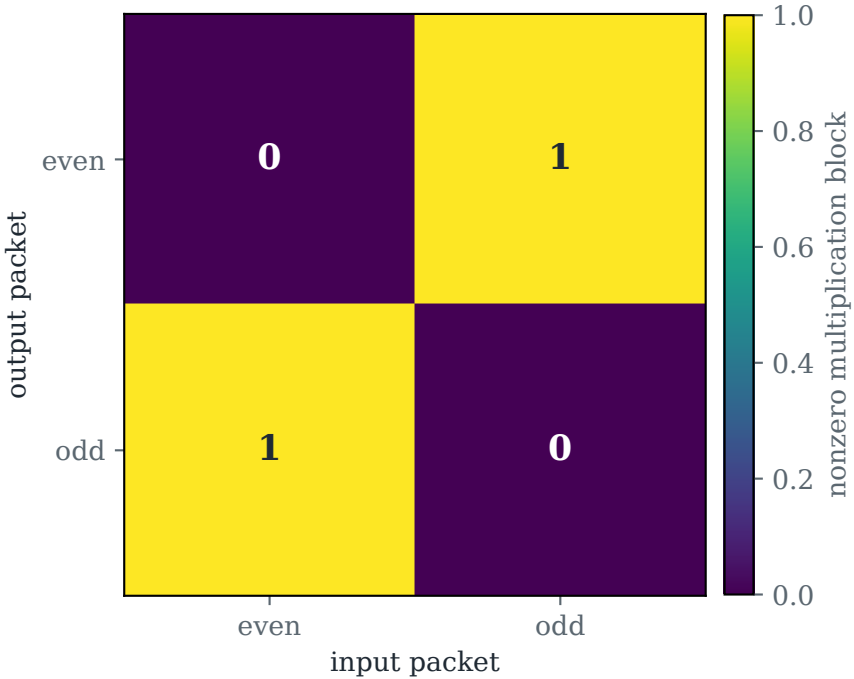
Packet transfer for multiplication by y

The operator exchanges the parity packets, while the square returns to the even side.



Coordinate ring

$k[x, y]/(y^2 - x^3 + x)$

even packet: $k[x]$

odd packet: $y\,k[x]$

M_y : even ↔ odd

$M_y^2 = M_{x^3 - x}$